

Rayan Morais

FULL-STACK SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with real products in production and a focus on justified architecture decisions. Strong command of TypeScript, Node.js, React, and Next.js. Hands-on experience with event-driven architectures (RabbitMQ), microservices with DDD (Domain-Driven Design), job queues (BullMQ/Redis), OAuth2/OIDC authentication, payment gateways (Stripe, PayPal, MercadoPago), and cloud deployment (AWS Lambda, Vercel, Railway). I value code that survives production: explicit error handling, data-driven refactoring, and clear documentation. Prior background in healthcare and law instilled zero tolerance for error and sharp attention to edge cases.

PROFESSIONAL EXPERIENCE

Software Engineer

Mar 2026 – Present

Vasc-Review — continuing medical education platform for vascular surgery residents and surgeons
vascreview.com.br

- Front-end web and mobile development (React Native):** new version of the platform featuring video lessons and a question bank for vascular surgery residency and board qualification — app published on the App Store and Play Store, with a new version under active development.
- Real production environment:** feature delivery, bug fixing, and maintenance of code used by medical specialists, under a formal code review process.

Freelance Full-Stack Developer

Jan 2025 – Present

Independent projects & team collaboration

- Collaboration with a Senior Software Engineer** (currently at AT&T) on full-stack projects — pair programming, code review, and feature delivery in a real production environment.
- Production bug diagnosis and resolution:** identifying N+1 queries, race conditions in upserts, and authentication vulnerabilities, plus third-party API integrations (affiliate networks, OAuth, payments, email) with explicit rate limit and retry handling.

Front-End Developer

Jan 2024 – Jan 2025

Apprenticeship program in software development

- Built React applications alongside a product team, focused on component design and API integration.

Nursing Intern

Jan 2022 – Dec 2024

HPS — Juiz de Fora, MG, Brazil

- Clinical practice under strict protocols, with direct responsibility for patient care in multidisciplinary teams — the origin of the process discipline and zero tolerance for error I apply to software today.

Legal Intern

2016 – 2018

Law firm

IT Support Analyst

2011 – 2014

City Council of Madre de Deus – MG, Brazil

- Managed network and hardware infrastructure in an institutional environment, handling the full lifecycle of technical support tickets.

TECHNICAL SKILLS

Languages

TypeScript · JavaScript (ES6+) · Go (Golang – Intermediate) · SQL · Python (basic)

Frontend

Next.js 15 · React 19 · React Native · Tailwind CSS · Framer Motion · Styled Components · Radix UI

Backend

Fastify · NestJS (Controllers/Services) · Node.js · Express · Go/Gin · BullMQ · RabbitMQ · Redis

(Queues/Workers) · Socket.io

Databases

PostgreSQL · MongoDB · MySQL · Supabase · Prisma ORM · Meilisearch

Auth & Payments

OAuth 2.0 (Open Authorization) / OIDC (OpenID Connect) · NextAuth v5 · Keycloak · JWT · Stripe · PayPal ·

MercadoPago

Cloud & DevOps

AWS Lambda · AWS SAM · AWS Secrets Manager · Docker · Vercel · Railway · GitHub Actions (CI/CD) · Kong

API Gateway

Architecture, Patterns & Quality

SOLID · DDD (Domain-Driven Design) · Clean Architecture · Event-Driven Architecture (EDA) ·

Microservices · Serverless · Controllers, Services & Workers (NestJS) · TDD (Vitest/Jest/Testing Library) · Clean Code · Design Tokens & Dynamic Componentization

Services

Cloudinary · Resend · Sentry · Anthropic Claude (API) · Pix (MercadoPago)

Design & Tooling

Figma (High-Fidelity) · React Hook Form + Zod · ESLint · Prettier · Husky · Knip · Git · Postman · Linux

FEATURED PROJECTS

CupomManiac — Discount Coupon Platform

2026

cupommaniac.com.br · github.com/rayancmorais/cupomManiac

Full-stack coupon platform for the Brazilian market (similar to Méliuz), built from scratch to production on its own domain.

- N+1 refactoring in the affiliate sync:** each cycle processed thousands of coupons (Lomadee, Admitad) as individual upserts, locking the database under load. Rewritten as upsertMany with batch deduplication by (source, externalId), eliminating thousands of round-trips per cycle, with parallel indexing into Meilisearch.

- **Neon pooler + PrismaClient singleton:** Neon's free tier caps direct connections at 10, and Next.js Server Components instantiating multiple PrismaClient instances exhausted the limit at peak. Migrated to the pooler endpoint with a singleton client on the backend.
- **Proxy routes to work around cross-origin cookies:** backend (Railway) and frontend (Vercel) on different domains caused SameSite=Lax to block cookies on fetch mutations. Sensitive mutations go through Next.js proxy routes that validate the session server-side before forwarding to the backend.
- **AlertMatcherService with O(1) round-trips:** naive alert matching would require N queries per new coupon. Rewritten as findMany → createMany (skipDuplicates) → updateMany — 3 batched calls per cycle, regardless of coupon volume.
- **5 affiliate sources via the Adapter Pattern:** Lomadee, Admitad, Awin, Amazon Associates, and Mercado Livre implement the same sync interface — adding a provider doesn't change the upsert service. Automated cron jobs run sync every 3h and expired-coupon cleanup daily at 2am.
- **Alert system with double opt-in and algorithmic scoring (0-100):** email verification → confirmation → batch matching → alert with token-based unsubscribe; coupons ranked by discount, code availability, and expiration proximity.

Stack: Next.js 15 (App Router, standalone output), TypeScript, Fastify, Prisma ORM, PostgreSQL (Neon), Redis (Upstash), BullMQ, Meilisearch, NextAuth v5, Resend, Sentry, Tailwind CSS, Vault, Vercel, Railway

Crash Game — Full-Stack Technical Challenge (Jungle Gaming)

2026

github.com/rayancmorais/fullstack-challengeRayancm

Real-time multiplayer crash game: the multiplier climbs from 1.00x and can crash at any moment — players bet and cash out before the crash.

- **Game Service and Wallet Service communicate exclusively via RabbitMQ** — no direct HTTP between services. Rationale: the Wallet must be the single source of truth for balances; synchronous HTTP would create coupling and inconsistency risk on partial failures.
- **DDD (Domain-Driven Design) with 4 layers:** the domain layer is pure TypeScript — zero NestJS, zero Prisma — so business rules are testable without framework mocks. 91 unit tests + 11 E2E scenarios covering bet→cashout→balance, bet→crash→loss, and limit validations.
- **Monetary values as BigInt cents (PostgreSQL BIGINT, TypeScript BigInt):** R\$1.50 = 150n — eliminating floating-point errors in financial calculations. DTOs serialize as strings to work around JSON.stringify limitations.
- **Provably fair with HMAC-SHA256:** the server seed hash is exposed before bets and the seed revealed after the crash — players can independently recompute the crash point and verify no manipulation occurred.
- **Kong API Gateway (DB-less)** for rate limiting and REST routing; WebSocket connects directly to the Game Service (port 4001) — DB-less Kong has limitations with HTTP→WS upgrades. CI/CD via GitHub Actions with build, test, lint, and Docker build-check before merge.

Stack: NestJS, TypeScript, Bun, Next.js 15, PostgreSQL, Prisma 7, RabbitMQ, Kong, Keycloak, Socket.io, Zustand, TanStack Query, Docker, GitHub Actions

Lux Lab Brasil — Dropshipping Platform (BR + EU)

2025

luxlabbr.vercel.app (BR version) · github.com/rayancmorais/luxeLab (EU version) · BR repo private

Multi-market full-stack dropshipping platform (Brazil and European Union), with international suppliers, multiple payment gateways, i18n, and an AI assistant.

- **Multi-supplier with geographic routing:** a single Supplier interface implemented by BR and EU providers (BigBuy, Griffati, Printful); PreferredForCountry(country) returns the list ordered by delivery country — adding a supplier doesn't change the order flow. Parallel catalog sync via goroutines.
- **Region-aware pricing pipeline:** supplier price normalization to the local currency (BRL or EUR), gross margin calculation, and destination-country tax rates (IVA/VAT) — three isolated, independently testable stages.
- **Multi-provider OAuth2/OIDC with PKCE:** Google, Facebook, and Apple — state signed via OAUTH_STATE_SECRET. Dual middleware (jose.jwtVerify on Next.js edge, golang-jwt in Go) with user/admin/worker roles mapped in both layers.
- **Streaming AI chat (SSE) under Lambda constraints:** API Gateway drops connections at 29s, making long streams unviable. Claude (Anthropic) chat sessions capped at ≤28s with an explicit handler timeout, documented for future migration to WebSocket API.
- **Production secrets hardening:** after an accidental credential exposure in .env, added gitleaks to CI to block commits containing secrets, with immediate rotation and migration of sensitive variables to AWS Secrets Manager.

Stack: Go, Gin, MongoDB Atlas, Next.js 15, TypeScript, Tailwind CSS v4, Radix UI, React Hook Form + Zod, AWS Lambda, AWS SAM, AWS Secrets Manager, Stripe, PayPal, MercadoPago, Cloudinary, Resend, next-intl, Claude (Anthropic), GitHub Actions

Personal Portfolio

2025

rayancmorais.com.br · github.com/rayancmorais/myportfolioofficial

- **Design system via CSS tokens:** the entire visual layer is driven by CSS custom properties (--cy, --fg-1, --bg-1...); dark/light theme switching via a data-theme attribute on <html>, with no component re-renders.
- **Smooth scrolling without recreating the loop on every render:** the useLenis hook instantiated Lenis inside the component — every re-render recreated the scroll loop, causing jank. Fixed with a module-level singleton that survives re-renders.
- **Fluid parallax with zero bundle cost:** hand-built with getBoundingClientRect + requestAnimationFrame instead of a ready-made library, with useReducedMotion() respected in every animation.

Stack: React 19, TypeScript, Vite, Styled Components, Framer Motion, Lenis, i18next, Vitest, Testing Library, ESLint, Prettier, Husky, Knip, Vercel

EDUCATION

B.Sc. in Software Engineering — UNINTER

In progress

B.Sc. in Nursing — UNIVERSO, Juiz de Fora, Brazil

Completed